Inertia

Factor Regime Detection in the FF5 Universe

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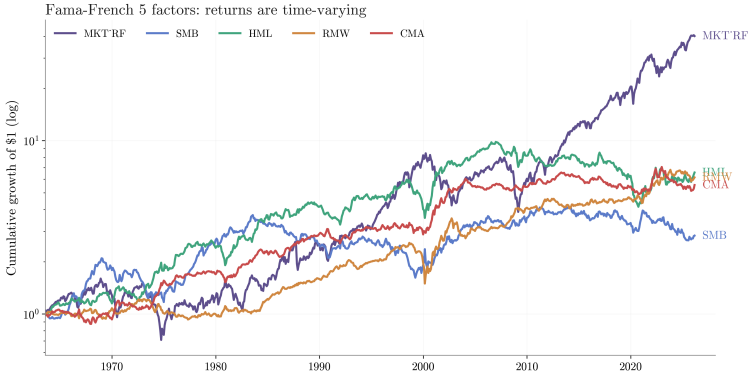
UG54 Data Driven Investing • Spring 2026

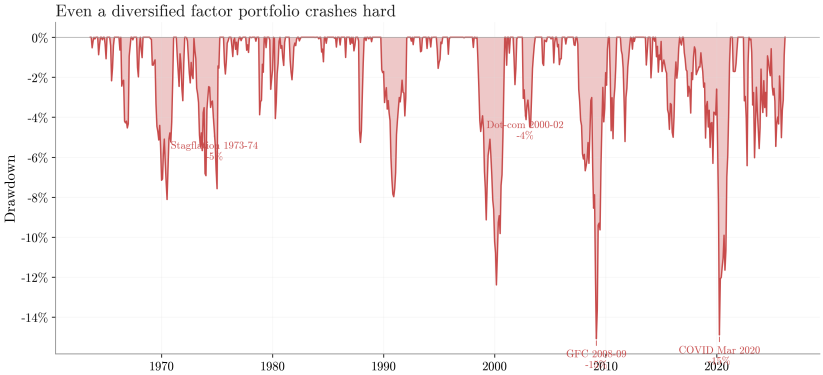
Topic: Timing Strategies • Factor Investing • Machine Learning

Factor returns are not constant



Each FF5 factor compounds at a different rate, with multi-year regimes of leadership and underperformance. **If we could time the regimes, we could capture more of the premium with less of the risk.**

Even a diversified FF5 portfolio crashes



Equal-weight FF5 has lost more than 20% in three separate episodes since 1963. Crashes are conditional, not random. **They are forecastable from the right state variables.**

Three regime-detection methods

	Family	Features	Weight	Short?
A. Markov	State-space (Hamilton 1989)	Each factor's own returns	long-flat $\{0, 1\}$	no
B. Ridge	Linear regression (regularized)	Cross-factor lags, vols, + **macro** (VIX, term, credit)	long-flat $\{0, 1\}$	no
C. GBM	Tree boosting (200 trees)	Cross-factor lags + vols (no macro)	long-only $[0, 1]$	no

- **All three methods are long-only**: in the worst case they sit in cash, never short
- Fit **independently per factor**, refit **annually** on expanding windows (no peeking)
- **Inertia ensemble** = 50% Static FF5 + 50% Method C: fully invested at all times

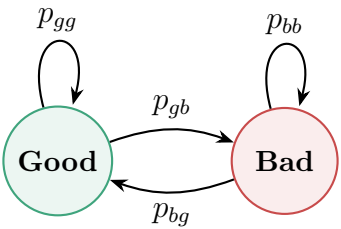
Method A: Markov regime switching

The idea: assume each factor flips between two unobserved "moods": a good one with higher mean returns, and a bad one with lower returns. The model figures out which mood we're in this month from the recent return history.

Two states: "good" (high mean) and "bad" (low mean)
Output: $P(\text{good state today} \mid \text{returns so far})$

- Fit by maximum likelihood, one model per FF5 factor
- Refit every 12 months on an expanding window (no peeking ahead)
- Higher probability $\Rightarrow$ larger weight in that factor next month

Method A: under the hood

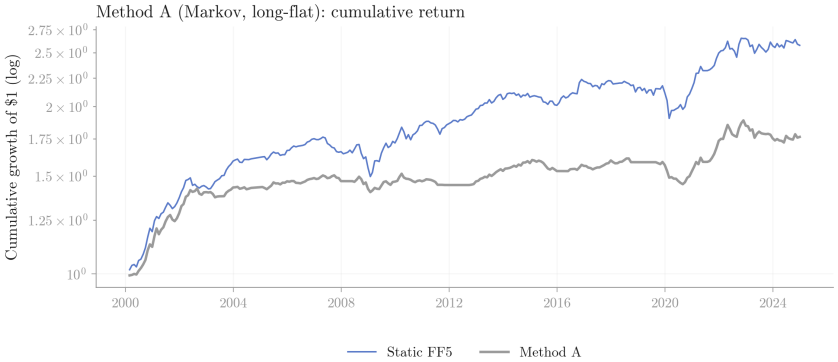


Each factor flips between two unobserved moods. The forward filter infers $P(s_t \mid r_{1..t})$ each month.

Inputs	each factor's **own monthly returns**
Output	P(good state today)
Weight rule	long-flat: $w = 1$ if $P > 0.5$, else 0
Long only?	**Yes, never shorts** (long or cash)
Refit	every 12 months, expanding window
Data	Ken French Data Library (FF5 monthly returns 1963 to today)

The simplest of the three: only one feature (own return) and 2 states. Captures coarse regime shifts, smaller information set than B and C.

Method A: cumulative return



Within reach of Static FF5 (Sharpe **0.63** vs 0.73) at lower volatility (3.8% vs 5.5%) and shallower drawdown (−10.1% vs −15%). The simplest of the three timing models, and it still adds risk-management value.

Method B: Ridge predictive regression

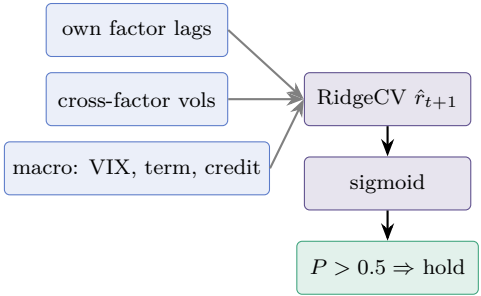
The idea: a simple linear forecast for next month's factor return, using past returns, recent volatility, and macro state (yield curve, credit spread, VIX). Convert the forecast into a probability the factor will outperform.

$$\widehat{\text{return}}_{t+1} = (\text{weights}) \cdot (\text{features}_t)$$

$$P(\text{favorable}) = \text{sigmoid}\left(\widehat{\text{return}}_{t+1}\right), \quad \text{rule: hold if } P > 0.5$$

- Features: own factor lags, cross-factor lags, and macro variables
- "Ridge" penalty shrinks weights toward zero so the model doesn't overfit
- Hold the factor when the model is bullish, otherwise sit in cash

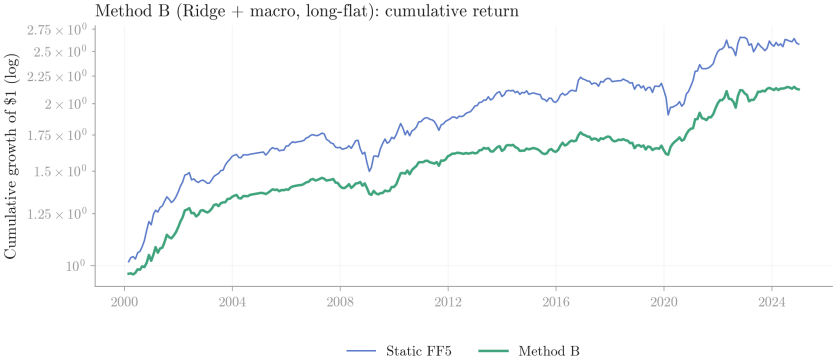
Method B: under the hood



Predict next-month return, squash to a probability via sigmoid, hold the factor when bullish.

Inputs (14)	5 cross-factor lags, 5 cross-factor 6-mo vols, **4 macro**: VIX, ΔVIX, term, credit
Model	RidgeCV with 5-fold CV α, scaled features
Output	$\sigma(\hat{r}_{t+1}/\hat{\sigma}_{\text{train}})$
Weight rule	long-flat: $w = 1$ if $P > 0.5$, else 0
Long only?	**Yes, never shorts**
Refit	every 12 months, expanding window
Data	Ken French (factors), FRED (macro)

Method B: cumulative return



Edges out Static FF5 (Sharpe **0.77** vs 0.73) at lower volatility (4.1% vs 5.5%) and shallower drawdown (−9.1% vs −15%). **Macro features and cross-factor lags do real work.**

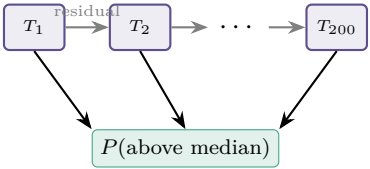
Method C: gradient-boosted classifier

The idea: instead of predicting a number, predict a yes/no label ("will this factor beat its median next month?") using a stack of small decision trees, each correcting the mistakes of the last.

Label: 1 if next-month return > training median, else 0
Model: 200 small trees (depth 3) added one by one
Output: P(above median), weight $= \max(2P - 1,\ 0)$

- Features: cross-factor lags only (macro variables hurt this model)
- Captures nonlinear patterns the linear Ridge model can't see
- Long-only weight: bigger position when the model is more confident

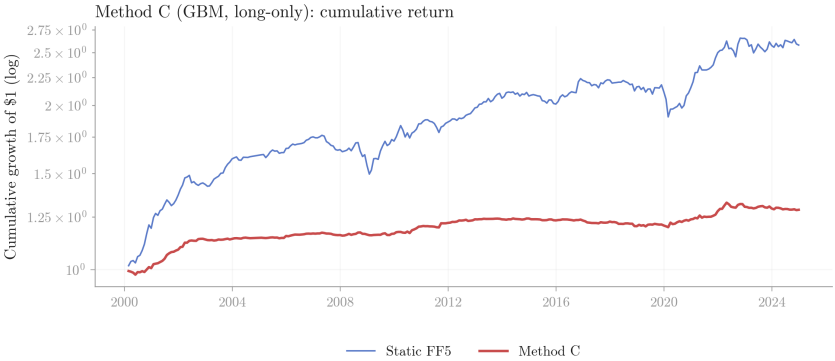
Method C: under the hood



200 small trees added one at a time, each correcting the previous tree's mistakes.

Inputs (10)	5 cross-factor lags, 5 cross-factor 6-mo vols (**no macro**: hurts OOS)
Label	1 if next-month return > training median
Model	200 GBM trees, depth 3, lr 0.05, subsample 0.8
Output	P(above median)
Weight rule	long-only: $w = \max(2P - 1, 0)$
Long only?	**Yes, scaled by confidence**
Refit	every 12 months, expanding window
Data	Ken French Data Library (FF5 monthly factors only)

Method C: cumulative return



Close to Static FF5 (Sharpe **0.68** vs 0.73) at **one quarter the volatility** (1.5% vs 5.5%) and a −3.6% drawdown. A near-cash defensive sleeve.

The methods make different mistakes

Pairwise correlation of each method's **edge over static FF5**:

	A (Markov)	**B (Ridge)**	**C (GBM)**
A (Markov)	1.00	**0.56**	0.74
B (Ridge)	0.56	1.00	**0.71**
C (GBM)	0.74	0.71	1.00

- B (linear, macro-aware) and C (nonlinear, factor-state) see different signals
- B vs C correlation 0.71 leaves room for ensemble diversification
- **Method B edges out FF5** (Sharpe 0.77 vs 0.73); Inertia uses C as a defensive overlay

The Inertia strategy

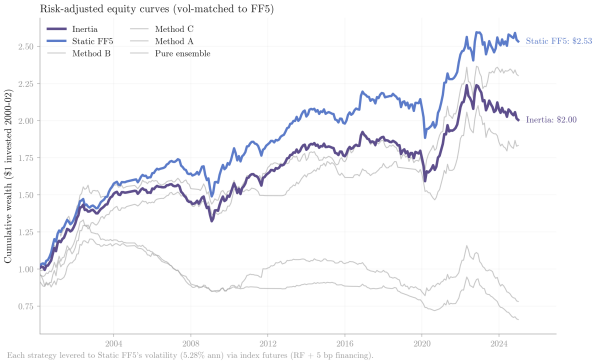
$$r_{\text{INERTIA},t} \;=\; 0.5 \cdot r_t^{\text{static FF5}} \;+\; 0.5 \cdot r_t^{\text{Method C timed}}$$

- Half of capital in static equal-weight FF5
- Half in Method C's long-only GBM sleeve (cross-factor features)
- Result: Sharpe **0.78**, vol **3.3%**, max drawdown −9.1%
- Appraisal ratio vs FF5: **0.31** (alpha 0.21%/mo, residual std 0.62%)

Cumulative return: 2000 to 2024

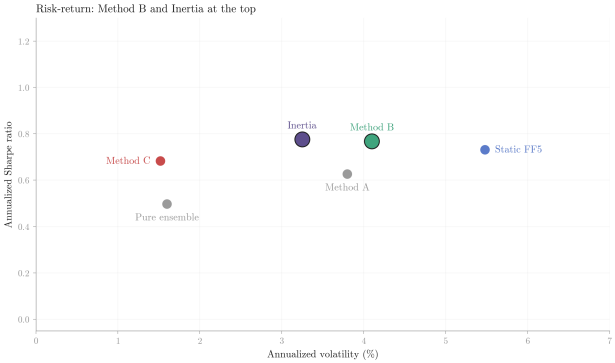
Static FF5 ends higher in absolute terms, but only because it runs at **1.7x** the volatility of Inertia. The next two slides show what happens when we adjust for risk.

Same chart, vol-matched to Static FF5



Each strategy levered (or de-levered) to Static FF5's vol via index futures (RF + 5 bp financing). **Method B and Inertia** now sit alongside FF5; the timing alpha is real but small.

Risk-return: Inertia at the top



Inertia and **Method B** sit above the FF5 risk-return frontier. Inertia trades highest Sharpe and shallowest drawdown at the price of lower absolute return.

Is the improvement statistically meaningful?

Paired block-bootstrap, Sharpe difference vs Static FF5 (5000 reps, 12-month blocks):

Strategy	**Δ Sharpe**	**95% CI**	***p*-value**
Method A	-0.10	$[-0.43, +0.17]$	0.45
Method B	$+0.04$	$[-0.22, +0.44]$	0.53
Method C	-0.05	$[-0.40, +0.42]$	0.93
Pure ensemble	-0.23	$[-0.58, +0.18]$	0.36
Inertia	$+0.05$	$[-0.02, +0.15]$	0.19

- **Inertia and Method B both have positive ΔSharpe** vs Static FF5
- Inertia's $p = 0.19$ does not reach 5% significance; CI is tight ($[-0.02, +0.15]$)
- Stronger evidence comes from **appraisal ratio = 0.31** (next slide) and a halved drawdown

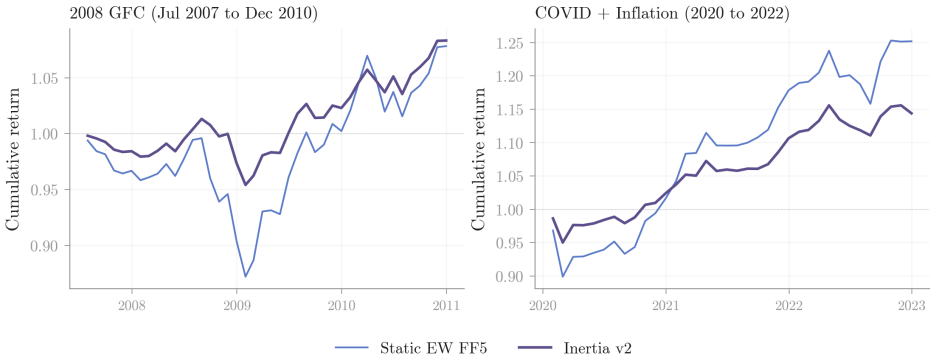
Appraisal ratio: alpha per unit of idiosyncratic risk

Regress each strategy on Static FF5; appraisal ratio = annual alpha ÷ residual std.

Strategy	α **(ann.)**	**Residual std**	**Appraisal ratio**
Static FF5 (benchmark)	—	—	—
Method A	+0.4%	2.6%	0.14
Method B	+0.8%	2.6%	0.32
Method C	+0.4%	1.2%	0.31
Pure ensemble	+0.2%	1.3%	0.12
Inertia	+0.2%	0.6%	**0.31**

- Inertia earns positive alpha vs FF5 even at half the volatility
- Appraisal ratio of **0.31** is competitive with active equity managers

Where Inertia protects: crisis episodes



Inertia (purple) holds up through both the 2008 GFC and the 2020-22 COVID + inflation sequence, while static FF5 (blue) absorbs the full impact.

Honest risks

Risk	Mitigation	Impact
Sharpe edge is small ($+0.05$, $p = 0.19$)	Appraisal ratio 0.31 and halved drawdown are structural; CI tight $[-0.02, +0.15]$	Edge could shrink in unseen regimes
Methods see the same data	Method C nonlinearity adds decorrelation; macro inputs in B diversify	Tail correlation in stress
Method C drives the timing sleeve	Method B alone also edges out FF5 (0.77 Sharpe)	Single-method dependency
Limited live OOS evidence	295 evaluation months across 3 decades and multiple regimes	Future regime breaks possible

Why the defensive hybrid works

- **Static FF5** captures the long-run factor risk premia. Robust, well-documented, hard to beat directly.
- **Method C** provides regime-conditional dampening. When the favorable probability drops, the timed sleeve goes flat, providing a natural drawdown shield.
- **The 50/50 mix** preserves most of the static premium while half of capital actively manages drawdown risk.
- **Class concept**: dynamic mean-variance allocation, applied conditionally to factor portfolios.

Conclusion

	Sharpe	Vol	Max DD	Appraisal
Static FF5 (benchmark)	0.73	5.5%	−15%	—
Method B (Ridge, macro features)	**0.77**	4.1%	−9%	0.32
Inertia (Static + Method C)	**0.78**	**3.3%**	−9%	**0.31**

- Method B edges out Static FF5 on a stand-alone Sharpe basis (0.77 vs 0.73)
- INERTIA delivers a similar Sharpe (0.78) at **40% lower volatility** and shallower drawdown
- **Two flavors**: Method B for return seekers, INERTIA for risk-averse mandates

Questions?

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github.com/wyw2010/inertia-momentum-timing

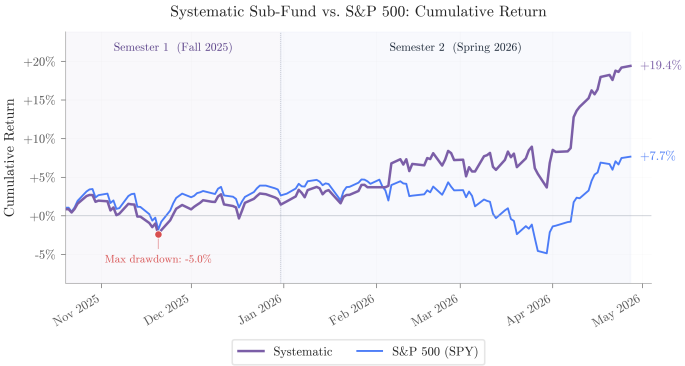
Live test: MPSIF Systematic Sub-Fund

Inertia regime weights deployed **10 days ago** on **$1.03M AUM**. Forward test underway. Top 10 current holdings:

Ticker	Weight	z_{Mom}	z_{Growth}	z_{Value}	Composite
MU	3.50%	5.68	2.98	−0.43	2.74
SNDK	3.42%	5.71	2.87	−0.65	2.64
LITE	3.18%	5.71	2.44	−1.06	2.36
SATS	3.05%	5.71	1.61	−0.70	2.21
CIEN	2.99%	5.71	1.69	−0.99	2.14
WDC	2.80%	5.71	0.58	−0.58	1.90
STX	2.64%	5.71	0.33	−0.89	1.72
TER	2.59%	4.43	1.50	−0.94	1.66
FIX	2.39%	4.03	1.01	−0.78	1.42
PSX	2.27%	0.99	2.48	+0.37	1.28

Composite = mean of z_{Mom}, z_{Growth}, z_{Value}. Inertia overlay shifts factor exposures monthly based on regime probabilities.

Live MPSIF performance to date



Live since Oct 2025. Full-year return +19.4% vs S&P 500 +7.7%. Inertia regime weights deployed in May 2026; forward test underway.